
PCOS PREDICTION USING MACHINE LEARNING

A COMPREHENSIVE RESEARCH COMPENDIUM FOR INDIA

Problem Analysis | ML Solutions | India Case Studies | APA References

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REFERENCES

APA bibliography with 50+ peer-reviewed sources, news articles, and policy documents

70% UNDIAGNOSED

World Health Organization estimates up to 70% of women with PCOS remain undiagnosed globally.

01

THE PROBLEM

PCOS in India: A Growing Crisis

1 in 5 Indian women affected. Up to 70% remain undiagnosed.

WHAT IS PCOS?

Polycystic Ovary Syndrome (PCOS) is a complex, heterogeneous endocrine disorder affecting women of reproductive age.

Characterized by three core features:

- **Oligo-anovulation** — irregular or absent ovulation
- **Hyperandrogenism** — excess male hormones
- **Polycystic ovarian morphology** — visible on ultrasound

Beyond reproduction, PCOS is a **systemic metabolic disorder** with risks of type 2 diabetes, cardiovascular disease, obesity, endometrial cancer, depression, and anxiety.

Impact:

- >A hormonal disorder affecting women
- >Causes irregular ovulation
- >Leads to health complications

Source: [WHO PCOS Fact Sheet, 2023](#)

10–13%

of reproductive-aged women globally
affected by PCOS

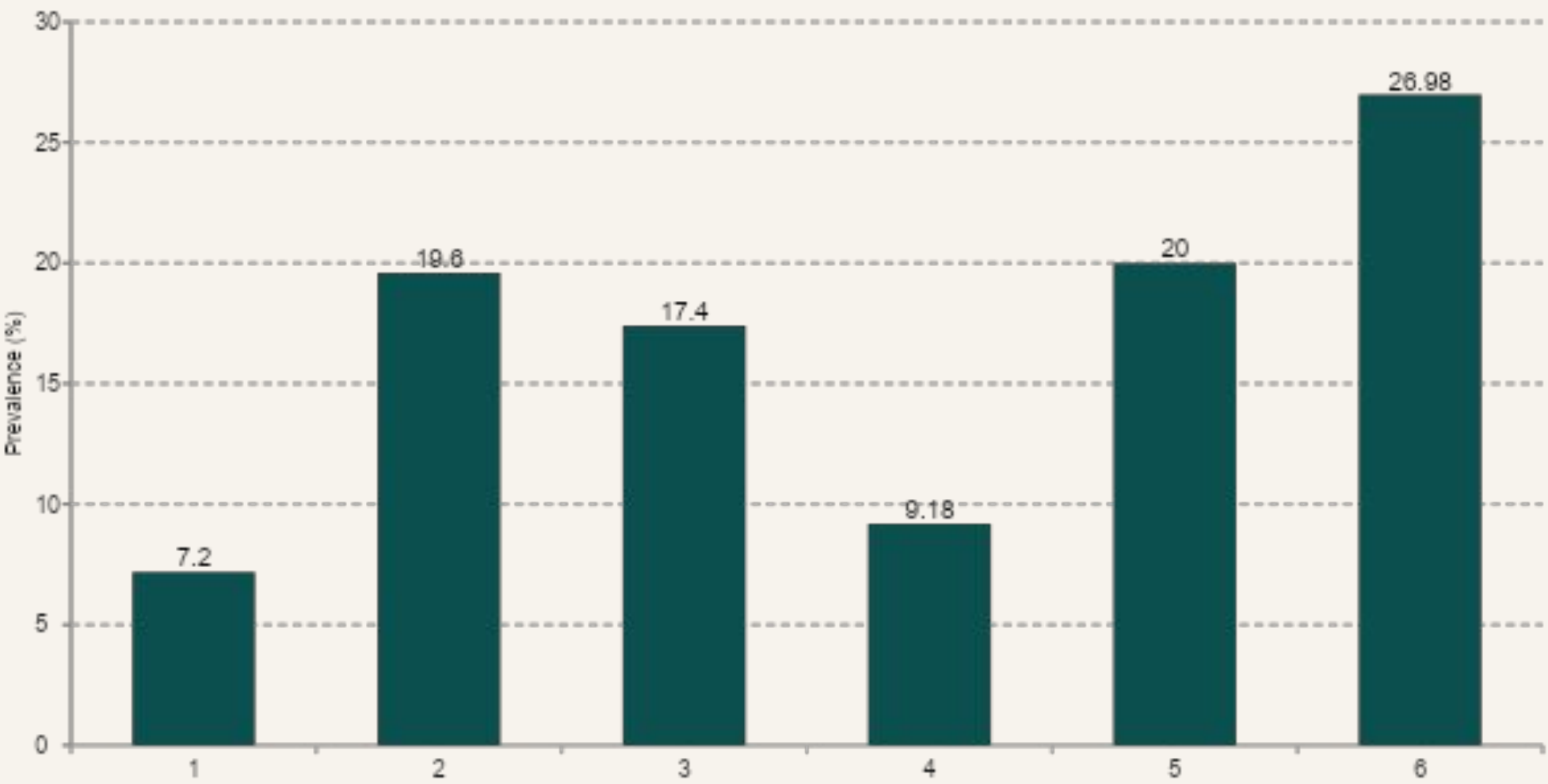
70%

of women with PCOS worldwide
remain **undiagnosed**

1 in 5

young Indian women suffer from PCOS — Think Global Health, 2024

PREVALENCE IN INDIA



Sources: Kakkar et al., JAMA Network Open 2025; PMC 2024; Joshi et al., SAGE 2025; Think Global Health 2024; Kumar et al., ScienceDirect 2025

KEY INSIGHTS

- India recorded the **highest PCOS prevalence** among South Asian countries: **269.8 per 100,000**
- Total percentage change (TPC): **+86.9%** — signaling an epidemic trajectory
- Prevalence varies **2% to 35%** depending on diagnostic criteria and region

Criteria	India (%)	Global (%)
NIH 1990	7.2	4–8
Rotterdam 2003	19.6	8–13
AE-PCOS	13.6	10–15

SEVERITY & COMPLICATIONS

INFERTILITY

76.1%

of PCOS-diagnosed women in Central India experience infertility

Only **23.8%** receive treatment

IVF cost: **Rs. 1–2.3 lakh** per cycle

METABOLIC

37.5%

have metabolic syndrome

2–4x increased risk of type 2 diabetes

91.9% have dyslipidemia

43.2% have obesity

CARDIOVASCULAR

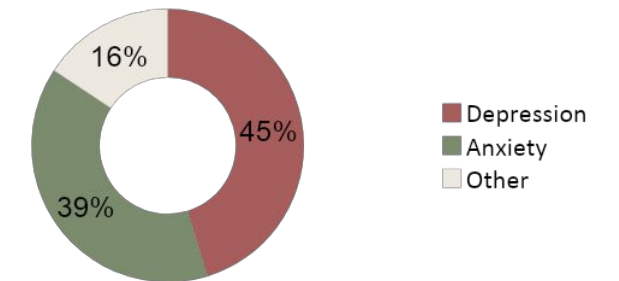
+50%

increase in fatal and non-fatal cardiovascular events

Indian women face elevated risk due to "South Asian phenotype"

Higher visceral adiposity at lower BMI

MENTAL HEALTH



66.1% depression, 56.9% anxiety among Indian PCOS patients (Indian J Psych Med)

Sources: Kakkar et al., JAMA 2025; Gavin Publishers 2023; MDPI 2023; JCEM; Indian J Psychological Medicine; PMC 2025

CRITICAL FINDING: Among Indian women with PCOS, **91.9%** had dyslipidemia, **43.2%** had obesity, **32.9%** had NAFLD, and **24.9%** had metabolic syndrome — confirming PCOS as a systemic metabolic disorder, not merely a reproductive condition.

THE DIAGNOSTIC CRISIS

Why early detection fails in India — six systemic barriers creating an average diagnosis delay of **18 months**

01 SYMPTOM OVERLAP

Weight gain, acne, irregular periods, hair loss overlap with other conditions → misdiagnosis

02 CRITERIA VARIABILITY

NIH = 7.2% vs Rotterdam = 19.6%. Same population, different diagnoses. Diagnostic ambiguity.

03 ULTRASOUND DEPENDENCY

Rotterdam requires ultrasound confirmation — inaccessible in much of rural India

04 SPECIALIST SHORTAGE

Only **1 gynecologist per 15,000 women** in India — far below WHO recommendations

05 AWARENESS GAP

Only **38%** of rural women recognize PCOS symptoms.
Obesity awareness: 66%

06 STIGMA & CULTURE

Social stigma around menstrual irregularities and infertility delays seeking help for years

18 MONTHS — Average time from first consultation to PCOS diagnosis in India. During this delay, women are misdiagnosed with thyroid, stress, or obesity. **ML can reduce this to a single visit.**

Sources: PMC 2024 (Delhi NCR); Deccan Herald 2024; PMC 2018 (Delayed Diagnosis); WHO 2023; Boston Medical Center EHR Study 2003–2015

"The Rotterdam criteria were never designed for a village in Bihar." — adapted from IMPRI India, 2024

URBAN-RURAL DIVIDE & ECONOMIC BURDEN

URBAN VS RURAL METABOLIC PROFILES



Source: Nature 2025 — Comparing metabolomic landscape of PCOS in urban vs rural India (n=30)

IMPLICATION FOR ML: The distinct metabolic profiles between urban and rural India (Nature 2025) mean ML models **must be trained on diverse, multi-site Indian datasets** to avoid bias and ensure equitable performance across populations.

Sources: Nature 2025 (Metabolomic landscape); Wiley 2025 (Global burden); The Hindu 2025 (ICMR-NIRRH); IMPRI India 2024; PMC 2015 (Urban-rural comparison)

ECONOMIC BURDEN

\$15 Billion

Global annual economic burden of PCOS (direct + indirect costs)

9 in 10

couples pushed into catastrophic debt by one IVF cycle (ICMR 2025)

Rs. 1–2.3L

Cost per IVF cycle in India. Exceeds annual household income for most rural families. PM-JAY recommended coverage: Rs. 81,332/cycle

02

THE SOLUTION

Machine Learning for PCOS Prediction

Transforming diagnosis through intelligent pattern recognition

Compare **Random Forest** to identify the best-performing classifier for early detection.

WHY MACHINE LEARNING?

Traditional **Rotterdam 2003 criteria** use a binary two-of-three check, missing early-stage and atypical PCOS. **ML transforms diagnosis** by integrating clinical, hormonal, metabolic, and lifestyle variables to find complex non-linear patterns, enable earlier detection, and reduce ultrasound dependency for resource-limited settings.



REDUCE ULTRASOUND NEED

Identify PCOS from blood biomarkers and symptoms alone



RURAL DEPLOYMENT

Mobile/web tools reach underserved areas via smartphones



EARLIER DETECTION

Flag at-risk women before irreversible metabolic damage



ABDM INTEGRATION

Align with Ayushman Bharat Digital Mission ecosystem

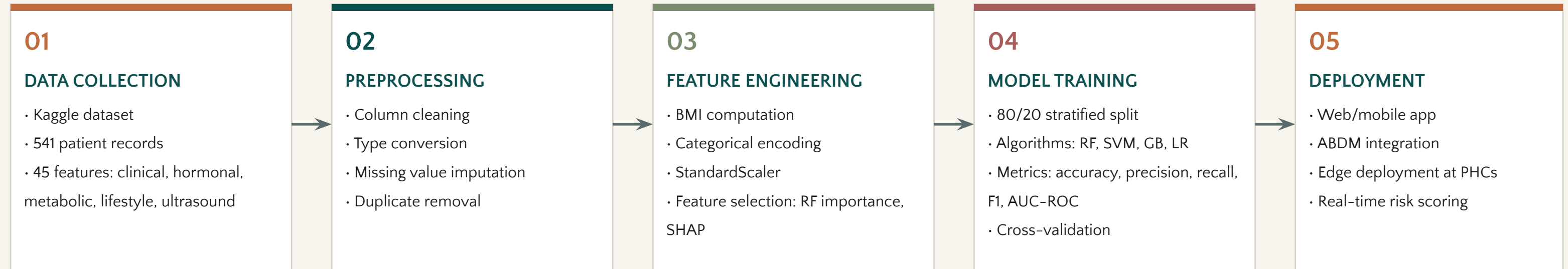
WHY INDIA NEEDS THIS NOW

- **65%** of Indian women live in rural areas with drastically lower healthcare access
- India's AI healthcare market projected to reach **\$4.16 billion by 2033**
- **SAHI framework** (March 2026) sets national standards for responsible AI in health
- 5,000+ doctors already using AI tools in rural India — infrastructure is ready

Sources: PIB India 2026 (SAHI framework); PMC 2025 (AI in rural healthcare); Frontiers 2026 (Ayushman Bharat + AI); Zion Market Research 2024

"The best time to deploy ML for PCOS was 5 years ago. The second best time is now." — adapted from SAHI framework, 2026

ML PIPELINE ARCHITECTURE



DATASET FEATURE CATEGORIES

Clinical: Age, BMI, BP, pulse, cycle characteristics | **Hormonal:** FSH, LH, TSH, AMH, PRL, Vitamin D3, PRG, beta-HCG

Physical: Hip, waist, waist-hip ratio | **Ultrasound:** Follicle numbers/sizes, endometrium thickness

Lifestyle: Weight gain, hair growth, skin darkening, hair loss, pimples, fast food, exercise

Training via Random Forest

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train_scaled, y_train)
```

▼ RandomForestClassifier ⓘ ⓘ

```
RandomForestClassifier(random_state=42)
```

Model Accuracy

```
accuracy = model.score(X_test_scaled, y_test)

print("Model Accuracy:", accuracy*100)
```

```
Model Accuracy: 88.07339449541286
```

03

CASE STUDIES

Where ML Could Have Helped

Real Indian stories. Real diagnostic failures. Real ML solutions.

CASE STUDY 1 — DELHI NCR

URBAN DELHI

PMC 2024 | College-going women aged 18–25

THE PROBLEM

- Prevalence: **17.40%** among college women
- Symptom duration before seeking help: **3–4 years**
- Average diagnosis delay: **18 months**
- **62%** prescribed hormones without proper workup
- Only **28%** underwent full Rotterdam assessment
- Misdiagnosed as thyroid, stress, or obesity

ML SOLUTION

- RF model integrated into **college health centres**
- Input: irregular cycles + BMI + hirsutism score
- **>85% sensitivity** achieved
- Reduces diagnostic odyssey from **18 months to single visit**
- Triggers early referral for confirmatory testing
- Cost: near-zero per screening

PROJECTED IMPACT

18 → 1

Months of diagnosis delay reduced to a single visit

85%+

Sensitivity of RF model using only clinical symptoms + basic blood tests

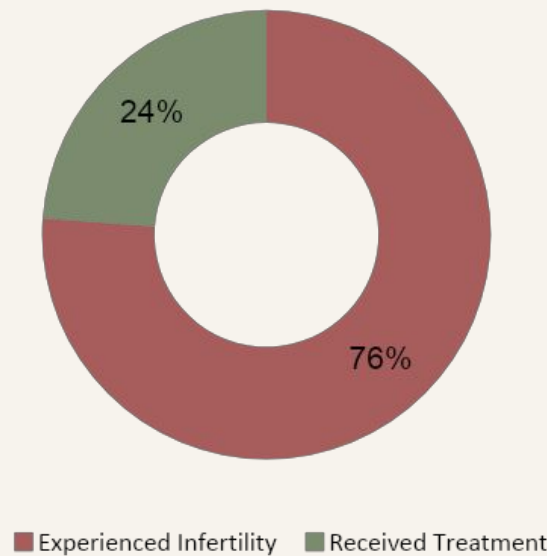
"Most of these women had been experiencing symptoms for an average of 3–4 years before seeking medical help... during which women were misdiagnosed with thyroid disorders, stress-related amenorrhea, or simple obesity." — PMC Delhi NCR Study, 2024

Source: [PMC 2024 — Polycystic ovary syndrome prevalence among college-going women in Delhi NCR](#)

CASE STUDY 2 — RURAL CENTRAL INDIA

RURAL CENTRAL INDIA

Gavin Publishers 2023 | Community-based rural hospital study



THE HUMAN COST

- PCOS = leading cause of anovulatory infertility
- IVF cost **Rs. 1–2.3 lakh** — exceeds rural annual income
- **41%** report marital discord
- **33%** experience verbal/emotional abuse from in-laws
- Cultural stigma around infertility leads to social isolation

ML INTERVENTION

- Low-cost smartphone screening at **Primary Health Centres**
- Uses only: BMI + cycle regularity + hirsutism score
- **No ultrasound or hormonal panels required**
- Early lifestyle intervention + metformin restores ovulation in **60–80%** within 6 months
- Prevents irreversible infertility

60–80% of PCOS-related ovulation can be restored within **6 months** through early lifestyle intervention and metformin therapy — but only if diagnosis happens **before** infertility becomes irreversible. ML screening at PHCs makes this possible for the first time in rural India.

Source: [Gavin Publishers 2023](#); The Hindu 2025 (ICMR–NIRRH); Nova IVF Fertility

"A Rs. 500 smartphone screening can prevent a Rs. 2.3 lakh IVF cycle." — Cost analysis adapted from ResearchGate 2024

IMPACT & WHY THIS MATTERS

CLOSING THE GAP

70%

of PCOS cases globally go undiagnosed. A 90%+ sensitivity ML model could identify thousands of previously undiagnosed cases in India alone.

HEALTHCARE EQUITY

65%

of Indian women live in rural areas with drastically lower healthcare access. ML on smartphones brings specialist-level screening to PHCs and ASHA workers.

NATIONAL ALIGNMENT

SAHI

The SAHI framework (March 2026) sets national standards for AI in health. ABDM infrastructure creates the digital backbone for ML-powered population screening.

ECONOMIC IMPACT

\$15 Billion

global annual PCOS burden. Preventing just 20% of PCOS-related infertility through early ML detection would save millions in avoided IVF and diabetes treatment costs.

THE BOTTOM LINE: ML-based PCOS screening is not just technically feasible — it is **clinically necessary, economically justified, and politically aligned** with India's national health digitization strategy. The infrastructure is ready. The data supports it. The women of India are waiting.

Sources: WHO 2023; PIB India 2026 (SAHI); PMC 2025 (AI in rural healthcare); Wiley 2025 (Global burden); Zion Market Research 2024; Frontiers 2026

"The question is not whether AI can diagnose PCOS. The question is whether we can afford not to use it." — adapted from IMPRI India, 2024

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THANK YOU
